

WRIGHT AAF, GA, USA

WMO: 722090

Lat: 31.883N

Lon: 81.567W

Elev: 45

StdP: 14.67

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 03875

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	27.1	30.4	11.6	10.0	37.9	16.1	12.4	38.6	21.3	53.9	19.3	53.0	3.0	320	0.360

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.9	96.2	76.6	94.4	76.7	92.4	76.4	80.6	89.2	79.6	88.7	78.5	87.5	7.3	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.6	148.6	84.1	77.3	142.4	83.0	76.4	137.9	82.1	44.1	89.3	43.0	88.7	42.0	88.1	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
17.8	15.5	12.8	DB	22.2	99.0	3.3	2.0	19.8	100.4	17.9	101.5	16.0	102.6	13.7	104.1
			WB	19.9	82.0	3.2	1.5	17.5	83.1	15.6	84.0	13.8	84.8	11.5	85.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	67.5	50.3	54.3	59.7	66.4	74.1	80.4	82.3	81.9	77.7	68.8	58.0	54.9
	DBStd	13.12	9.65	9.03	8.23	6.60	5.16	3.57	2.94	2.81	3.86	6.93	7.91	9.02
	HDD50	263	118	54	17	1	0	0	0	0	0	0	19	54
	HDD65	1650	461	312	202	63	4	0	0	0	0	43	235	330
	CDD50	6640	127	175	320	492	747	913	1001	987	830	583	259	206
	CDD65	2552	6	13	39	104	286	463	536	522	380	161	25	17
	CDH74	23728	35	115	374	1059	2648	4542	5424	5008	3078	1208	158	79
	CDH80	10039	1	11	65	288	1068	2167	2609	2259	1232	322	14	3
Wind	WSAvg	4.9	5.2	5.5	5.9	5.8	5.3	4.9	4.6	4.2	4.2	4.5	4.5	4.6
Precipitation	PrecAvg	49.8	3.4	3.5	3.7	2.8	3.2	6.5	5.6	6.3	4.6	4.1	2.0	2.8
	PrecMax	63.4	11.0	9.1	9.6	7.0	7.9	10.4	9.9	12.8	11.6	13.9	5.4	7.7
	PrecMin	34.1	0.4	0.7	0.4	0.4	0.8	2.0	2.1	3.0	1.4	0.2	0.1	0.6
	PrecStd	8.7	2.4	1.9	2.1	1.7	1.9	2.4	1.9	1.9	2.3	3.5	1.5	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	78.0	81.7	85.2	89.1	97.0	97.9	98.5	98.0	95.1	89.5	82.2	79.8
		MCWB	66.5	67.3	65.5	69.2	71.5	75.4	77.5	78.3	74.6	75.1	69.7	70.6
	2%	DB	73.7	77.6	81.8	86.3	92.1	95.3	96.0	95.2	91.3	86.3	78.5	76.3
		MCWB	63.6	65.1	65.9	68.5	71.7	76.4	77.3	78.2	75.3	73.0	67.2	67.6
	5%	DB	69.5	74.1	78.3	83.6	88.8	93.1	94.0	92.8	89.3	83.5	75.3	72.9
		MCWB	61.5	62.9	64.5	67.8	71.3	76.2	77.2	77.9	75.2	71.2	65.8	66.4
	10%	DB	66.1	69.9	74.8	80.6	86.1	90.7	91.7	90.6	87.2	81.1	72.1	69.3
		MCWB	60.4	60.6	62.5	66.4	71.0	76.0	77.0	77.7	74.8	69.9	64.2	64.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.1	69.7	69.8	74.4	77.4	81.4	81.4	82.9	80.4	77.9	73.8	73.3
		MCDB	73.3	76.5	78.6	82.8	85.4	90.2	89.8	91.1	87.2	83.7	77.9	77.6
	2%	WB	66.7	67.4	68.2	72.5	75.4	79.8	80.0	81.1	78.1	76.2	71.3	70.3
		MCDB	70.9	74.6	76.7	80.2	83.4	89.8	89.8	89.1	85.5	81.5	74.3	73.2
	5%	WB	64.5	65.2	66.6	70.8	74.2	78.4	79.0	79.8	77.2	74.7	68.9	67.9
		MCDB	68.2	72.1	74.5	78.3	82.7	88.3	89.1	88.2	84.9	79.8	72.4	70.7
	10%	WB	61.4	62.9	65.0	68.9	73.0	77.4	78.1	78.7	76.3	73.2	66.0	65.5
		MCDB	65.4	67.8	72.4	76.5	81.8	86.9	87.9	86.9	83.4	78.3	71.3	69.8
Mean Daily Temperature Range	5% DB	MDBR	21.8	23.0	23.8	24.4	22.2	19.8	18.9	17.7	17.9	21.2	23.1	21.0
		MCDBR	24.8	26.0	26.5	26.9	25.8	22.7	21.4	20.5	21.2	22.2	24.0	21.9
		MCWBR	15.6	14.9	12.5	11.9	9.3	7.7	6.8	6.8	7.4	10.5	14.0	14.3
	5% WB	MCDBR	20.3	22.2	22.9	22.8	20.6	20.4	19.1	18.3	17.0	17.2	18.3	19.4
		MCWBR	15.2	14.6	12.2	11.8	8.7	8.2	7.3	7.3	6.7	9.3	13.1	14.8
Clear-Sky Solar Irradiance	taub		0.331	0.339	0.374	0.401	0.446	0.496	0.522	0.512	0.428	0.394	0.355	0.343
	taud		2.513	2.491	2.416	2.334	2.241	2.156	2.094	2.124	2.358	2.416	2.479	2.509
	Ebn at Noon		283	292	288	282	269	255	247	248	267	269	272	272
	Edh at Noon		28	31	36	40	44	48	51	49	37	33	28	26
All-Sky Solar Radiation	RadAvg		947	1139	1499	1871	2018	1973	1957	1771	1508	1318	1047	831
	RadStd		51	117	135	125	118	152	122	110	125	136	69	69

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (36 neighbors)	+1.01	+1.40	N/A	N/A	N/A	N/A	N/A	-151	+344	+239

Nomenclature: See separate page